Shanghai: CMSRN performance by routine-pattern deviation

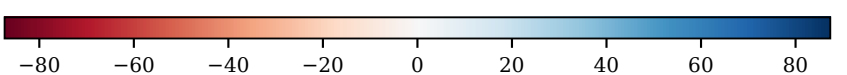
H6 (30 min) · full test set · training-fitted thresholds

(a) Inflow

	Low	Moderate	High	Extreme
HA	+82.7	+39.2	+12.3	+13.3
SVR	+50.1	+17.8	+8.4	+7.7
VAR	+60.5	+25.3	+10.8	+7.8
DCRNN	+10.1	+2.2	+1.9	+3.6
STGCN	+2.2	+1.2	+1.4	+4.0
GWNet	+1.2	+1.1	+1.6	+3.6
STNorm	-0.1	-1.1	-1.2	+0.5
DSTAGNN	+78.7	+33.0	+8.9	+5.1
PDFormer	+74.5	+52.5	+48.8	+57.7
STWave	-0.9	-1.0	-1.1	+0.2
PatchTST	+57.9	+6.4	+5.4	+10.2
TimeMixer	+58.9	+10.6	+8.2	+10.2
STDN	+84.6	+53.9	+25.6	+15.3
MetaDG	+83.1	+49.8	+24.8	+15.2
HyperD	+5.0	+0.8	+0.3	+2.8
RAST	+1.0	+0.9	+1.4	+3.1
CMSRN MAE	**0.177**	**0.536**	**0.851**	**1.296**
Valid n	92,199	223,697	222,486	115,210

(b) Outflow

	Low	Moderate	High	Extreme
HA	+84.4	+37.9	+12.1	+12.1
SVR	+53.6	+16.5	+8.1	+7.2
VAR	+64.9	+23.6	+10.8	+7.4
DCRNN	+11.1	+2.1	+1.6	+3.4
STGCN	+2.1	+0.7	+1.1	+3.0
GWNet	+0.8	+0.6	+1.1	+3.6
STNorm	+0.1	-1.2	-1.2	+0.4
DSTAGNN	+81.2	+32.1	+9.3	+3.6
PDFormer	+76.7	+52.2	+49.9	+58.8
STWave	-0.3	-1.1	-1.0	+0.2
PatchTST	+61.5	+6.0	+5.1	+9.5
TimeMixer	+62.2	+9.8	+8.1	+9.7
STDN	+85.7	+43.0	+14.0	+6.1
MetaDG	+84.8	+49.6	+28.0	+16.6
HyperD	+1.9	+0.4	+0.0	+1.1
RAST	+2.3	+1.2	+1.5	+4.1
CMSRN MAE	**0.158**	**0.553**	**0.870**	**1.480**
Valid n	84,381	216,920	227,734	110,618



CMSRN MAE improvement over each baseline (%)

Improvement = 100 × (MAEbaseline − MAECMSRN) / MAEbaseline. Blue: CMSRN better; red: baseline better.

Identical color scale across cities and the four-panel figure; no finite values are clipped.